Editor

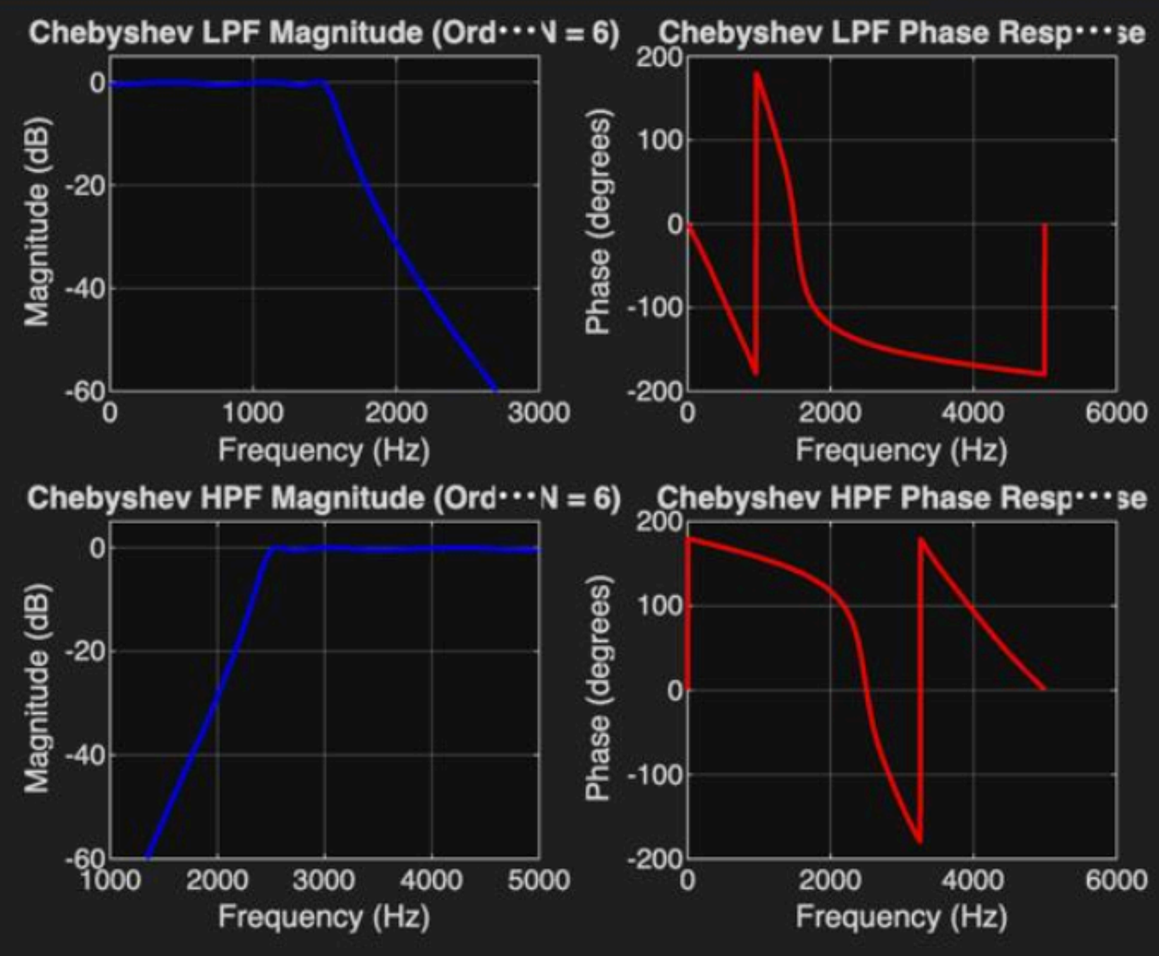
5g6g.m × | exp125g6g.m × | exp135g6g.m × | exp5g6g.m × | exp155g6g.m × | smart.m × | untitled.m × | ML.m × | ML2.m × | mlll.m × | mageshco1exp1.m × | MaheshCo1ex2.m ×

/MATLAB Drive/MaheshCo1ex2.m

```
clear; clc; close all;
% Design Parameters
Fs = 10000; Ap = 0.5; As = 30; % Sampling frequency in Hz
% Passband ripple in dB
% Stopband attenuation in dB
% 1. Low-Pass Filter Design
fp_lp = 1500; fs_lp = 2000; wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);
% Passband frequency (Hz)
% Stopband frequency (Hz)
[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');
% 2. High-Pass Filter Design
fp_hp = 2500; fs_hp = 1800; wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);
% Passband frequency (Hz)
% Stopband frequency (Hz)
[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');
% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===\n\n');
fprintf('--- CHEBYSHEV LOW-PASS FILTER ---\n');
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp);
fprintf('Passband Edge Cutoff Frequency (Wn_lp): %.4f Hz\n', Wn_lp * (Fs/2))
fprintf('Numerator Coefficients (b_lp):\n');
disp(b_lp);
fprintf('Denominator Coefficients (a_lp):\n');
disp(a_lp);
fprintf('--- CHEBYSHEV HIGH-PASS FILTER ---\n');
fprintf('Calculated Filter Order (N_hp): %d\n', N_hp);
```

Figure 1: Chebyshev Type-I LPF and HPF Responses

FIGURE FORMAT TOOLS LAYOUT



Command Window

New to MATLAB? See resources for Getting Started.

```
Passband Edge Cutoff Frequency (Wn_hp): 2500.0000 Hz
Numerator Coefficients (b_hp):
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117

Denominator Coefficients (a_hp):
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650

>>
```